

Hydrogen-based Hybrid Renewable Energy System Sizing Optimization using HOMER

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Abstract— Conventional power systems are mostly based on fossil fuels. These constitute a substantial source of pollution in addition to being quickly depleted. Isolated areas and communities cannot profit from this type of electric energy for the reason that they are far from the grid. The hybrid renewable energy systems (HRES) have demonstrated to be an excellent solution for such a situation. The purpose of this paper is to design and optimize a standalone hybrid energy system to satisfy the electricity demand for a remote drilling site in Adrar province located in the South of Algeria. To cure the problem of intermittence of the solar and wind sources, a hydrogen-based set up is appended to the hybrid renewable energy system. The simulations are done using HOMER software to determine the optimal the HRES configurations considering techno-economic criteria.

Keywords—Renewable Energies; Hydrogen System, Sizing; Techno-economic Study; HOMER

I. INTRODUCTION

Renewable energy technologies have received increased worldwide attention in recent years due to their favorable influence on resolving environmental problems (CO₂ emissions that cause global warming) and their potential to fulfill rising energy demand. Furthermore, the cheap operating and maintenance costs, as well as the convenience of installation, have led in the integration of renewable energy sources (RES) in recent years. Renewable energy has made significant technological advances as a clean source of energy with no fuel expense, with further continuing research in academia and business. Renewable energy accounted for over 62% of total installed power production capacity in 2016, representing a 9% increase over the previous year [1]. According to historical statistics from the International Energy Agency (IEA) [2] there is a strong indication of renewable energy dominance in power production. Targets to minimize greenhouse gas emissions (GHG), as well as government incentives to encourage RES, have promoted the greater integration of renewable energy technology in power networks. However, the current technological concerns about intermittent RES have restricted RES uptake in the existing electricity system.

Access to electricity has become a primary concern in rural areas due to geographical location, long distances from the main electric power grid, high cost of power transmission lines, technical constraints in remote locations, and reliability concerns of electric power supply due to natural adversities. Renewable energy resources, particularly Photovoltaic (PV) and wind, have the potential to be a useful solution for rural

power delivery [3,4]. This setup may offer a consistent power supply and may be more lucrative than standard electrification options for rural regions, notably in telecommunications, hazardous chemical companies in distant locations, mountainous places, and tiny islands, among other things.

In Algeria, residential buildings accounted for around 43% of total national electricity usage [5]. Building electricity consumption has increased quickly in most developing nations in recent years, owing to the use of novel technologies, the desire for entertainment, and the requirement for thermal comfort [6]. For example, Algerian demand for energy in the construction sector increased by 8% in 2018 compared to 2017 [7]. The fundamental issue with this increase is that in most nations, fossil fuels constitute the primary source of electricity generation [8] (more than 97% in Algeria) [9]. Furthermore, there are still a considerable number of buildings in isolated rural areas that do not have access to the energy grid. As a result, they must rely on diesel generators to supply their electrical needs. Because of the high cost of diesel fuel, which is likely to rise, and the health risks associated with it, utilizing diesel generators has become uneconomical [10] and contributes considerably to global warming, which has become the world's most pressing environmental issue [11]. Furthermore, the depletion of fossil fuel reserves is a significant challenge. As a result, incorporating renewable energy sources such as solar and wind is becoming increasingly important nowadays.

Given the fact that the government subsidizes diesel fuel, these costs have an effect on Algeria's economy, which is totally based on hydrocarbon income. Even though fuel costs are going up every day, conventional sources are hard to use because of environmental concerns and the fact that fossil fuels are running out. Because of this, it will be necessary to move toward using more renewable resources to make electricity. The country is lucky in that it has a huge amount of solar energy potential, which is more than 30 times the amount of energy the whole world uses in a year [12]. In this way, the Algerian government has put in place a number of policies and programs to encourage the use of renewable energy sources, increase the number of energy sources in the country, and make sure that even remote areas have access to electricity. Algeria made changes to its Renewable Energy and Energy Efficiency program in 2015. The program was first started in 2011. The updated version of the Program wants to get about 27% of all electricity produced from renewable sources [13].

Aside from that, the goal of this program is to cut GHG emissions by 7% by 2030 [14].

The Algerian government wants to develop solar energy more than any other renewable energy source. But wind, biomass, and geothermal power are not really big. In response to this program, the national company of oil drilling (ENTP) has set itself the goal of completing the Green Cabin project, which is a prototype of a self-sufficient Saharan cabin that runs on solar energy alone. But because this type of renewable electricity generation is unpredictable and only works sometimes, it throws off the supply and demand balance and cannot completely replace fossil fuels. So, a promising solution is the hybrid renewable energy system, which is made up of renewable sources, conventional sources, and/or an energy storage system. Recently, the hybrid renewable energy sources (HRES), especially solar-wind based hybrid renewable energy systems, have been used a lot to bring electricity to remote areas because they are more reliable and cost less than diesel-only systems [15]. Researchers face a big challenge, though, when they try to find the best way to size and set up HRES while taking into account many constraints. Many research papers on this topic have been written using optimization algorithms (particle swarm optimization, genetic algorithm, Ant colony algorithms) or widely used commercial software like HOMER, RETSCREEN, and iHOGA [16] [17].

The study focuses on the optimization and the design of a stand-alone (off-grid) HRES, and the modeling and simulation of the system with relevant data to determine its performance and contribution. The case study was picked based on the available potential of wind and solar energies. The primary goals of this work are to optimize the sizing of hybrid system components to reduce the cost of energy generation, as well as to maximize the utilization of solar panels while limiting pollutant emissions. This study aims also to advertize for hydrogen energy as a solution to the fluctuating nature of the solar and wind resources.

II. SPECIFICATION OF SYSTEM COMPONENTS

Modeling of energy system components is required to evaluate performance under various scenarios when using HOMER Pro for sizing optimization. The energy system components are mathematically modeled and their specifications presented in the following sections. The input parameters for each of the system components that need to be input to HOMER Pro are described in this section. These were used for optimizing the sizing of solar PV, battery storage, converter, wind turbine, diesel generator, fuel cell, electrolyzer, and hydrogen tank.

A. Solar PV Arrays

The PV array's power output (PPV) is calculated as:

$$P_{PV} = Y_{PV} \times x_{PV} \times \frac{G_T}{G_{T,STC}} \times [1 + \alpha_p \times (T_c - T_{c,STC})] \quad (1)$$

Where

x_{PV} : the PV derating factor (%).

Y_{PV} : the the rated capacity of the PV array (kW).

G_T : the solar radiation incident on the PV array in the current time step (kW/m²).

$G_{T,STC}$: the incident radiation at standard test conditions (1 kW/m²).

α_p : the temperature coefficient of power (%/°C).

T_c : the PV cell temperature in the current time step (°C).

$T_{c,STC}$: the PV cell temperature under standard test conditions (25 °C).

The derating factor is a scaling factor that is used to account for wire losses, dust on the panel, or anything else that could cause the output of the PV array to be different from what would be expected under rate conditions.

The solar radiation and the temperature are used to figure out the maximum power point, which is the voltage at which the array produces the most power. A maximum power point tracker (MPPT) is a solid-state device that is placed between the PV array and the rest of the DC components of the system to isolate the array voltage from the rest of the system and make sure that the array voltage is always equal to the maximum power point. HOMER Pro basically assumes the presence of an MPPT in the system by neglecting the influence of the voltage that the PV array is subjected to [18]. The temperature of the PV cells is referred to as the PV cell temperature. HOMER Pro estimates the cell temperature in each time step and uses it to determine the PV array's power output [19].

A generic flat-plate PV system was considered. Table 1 describes the PV system's input specifications.

Table 1 Specifications of PV System

Parameter	Value	References
Capital Cost(\$/kW)installed	1200	[20]
O&M Cost(\$/kW/year)	10	[20]
Replacement Cost(\$/kW)	1200	Estimated
Lifetime (years)	25	[20]
Efficiency (%)	15.5	[20]
Derating Factor (%)	88	HOMER Default
Maximum Capacity (kW)	224	HOMER Default

B. The battery bank

A battery bank is a grouping of one or more separate batteries. HOMER Pro models a single battery as a device that can store a certain amount of direct current (DC) electricity at a fixed round-trip energy efficiency (RTE), with limits on how quickly it can be charged or discharged, how deeply it can be discharged without being damaged, and how much energy can cycle through it before it needs to be replaced.

Excess power generated by renewable and/or a diesel generator is utilized to charge the battery bank, while energy shortages can be met by a battery bank or a diesel generator. Equation (2) [21] is used to calculate the state of charge of a battery while it is charging. Table 2 describes the battery's input specifications.

$$E_b(t + 1) = E_b(t) \times (1 - \sigma) + \left(E_g(t) - \frac{E_l(t)}{\eta_{inv}} \right) \times \eta_{BC} \quad (2)$$

In the discharging phase, however, the level of charge of the battery may be determined using Equation (3) [21].

$$E_b(t+1) = E_b(t) \times (1 - \sigma) + \left(\frac{E_l(t)}{\eta_{inv}} - E_g(t) \right) / \eta_{BD} \quad (3)$$

The level of charge of the battery $E_b(t)$ at each hour (t) is bound by a minimum and maximum storage capacity, E_{bmin} and E_{bmax} , as defined in (4)

$$\begin{aligned} & \text{If } E_{bmin} \leq E_b(t) \leq E_{bmax} \\ & \text{Then: } E_{bmin} = (1 - DOD) \times E_{bmax} \end{aligned} \quad (4)$$

Where

DOD: the depth of discharge

η_{inv} : the efficiency of the inverter

η_{BC} : the battery charge efficiency

η_{BD} : the battery discharge efficiency

σ : the battery's self-discharge, which is ignored here

E_g : the generated power

$E_l(t)$: the load at the hour t

Table 2 Specifications of Battery Bank

Parameter	Value	References
Type	Lithium-ion	Estimated
Capital Cost(\$/kWh)installed	350	[22]
O&M Cost(\$/kWh/year)	10	[22]
Replacement Cost(\$/kW)	350	Estimated
Lifetime(years)	15	HOMER Default
Round Trip Efficiency (%)	90	[22]

C. Bidirectional Converter

The bi-directional converter controls current flow in either direction (between AC and DC bus lines). The converter's primary role is to supply electricity from DC sources to the load. The size of the converter is determined by the system's maximum energy level [23]. The converter's capital cost and replacement cost are the same and are taken at \$115/kW [24]. This converter has an efficiency of 95 % and a lifetime of 15 years.

D. The Wind Turbine

Wind turbines (WT) are a free and readily available energy source that may be utilized to generate electricity by converting kinetic energy into electrical energy. This source is more efficient in areas with significant wind speeds, such as the southern portions of Algeria.

HOMER program calculates wind turbine power output based on wind speed data at the hub high and the manufacturer's wind turbine power curve. An 80 kW wind energy source (WES) wind turbine type was chosen for this purpose. The WT's capital and replacement expenses, as well as operating and maintenance expenditures, are \$1700/kW and \$15/kW/year, respectively [25]. The chosen WT has a 20-year lifetime with a hub height of 30 m.

E. Diesel Generator

The generator burns fuel to generate electricity, but it also generates heat, which may be utilized in Combined Heat and Power (CHP) applications. The generator's primary physical attributes are its maximum and minimum electrical power output, its estimated lifetime in running hours, the kind of fuel it uses, and its fuel curve, which connects the amount

of fuel spent to the amount of electrical power generated. HOMER Pro assumes the fuel curve is a straight line with a y-intercept and calculates the generator's fuel consumption rate F as:

$$F = F_0 \times Y_{gen} + F_1 \times P_{gen} \quad (5)$$

Where

F: the fuel curve slope

F₀: the fuel curve intercept coefficient

P_{gen}: the electrical output of the generator (kW)

Y_{gen}: the rated capacity of the generator (kW)

Diesel fuel prices in Algeria are currently at 0.19 \$/L and are anticipated to be at 0.65 \$/L by 2030 [26].

Table 3 Specifications of Wind Turbine

Parameter	Value	References
Capital Cost(\$)/installed	1700	[27]
O&M Cost(\$/op.hr)	15	[27]
Replacement Cost(\$)	1700	Estimated
Hub Height (m)	30	Estimated
Lifetime(year)	20	Estimated

Table 4 Specifications of Diesel Generator

Parameter	Value	References
Capital Cost(\$)/installed	500	[26]
O&M Cost(\$/op.hr)	0.030	[26]
Replacement Cost(\$)	500	Estimated
Minimum load ratio(%)	25	HOMER Default
Lifetime(h)	60000	Estimated

F. Electrolyzer

An electrolyzer uses electricity to produce hydrogen through water electrolysis. HOMER Pro defines the efficiency of an electrolyzer as the amount of energy in the hydrogen made (based on a higher heating value) divided by the amount of electricity used. A polymer electrolyte membrane (PEM) type electrolyzer is being investigated due to its technological maturity and reasonable efficiency under various loading situations. Table 5 describes the characteristics of the chosen electrolyzer.

Table 5 Specifications of Electrolyzer

Parameter	Value	References
Capital Cost (\$/kgH ₂) installed	1500	[28]
O&M Cost(\$/kgH ₂)	10	[29]
Replacement Cost(\$/kW)	1500	Estimated
Lifetime (years)	20	[30]

G. Hydrogen Storage Tank

The hydrogen tank stores the hydrogen produced by the electrolyzer for later use in a hydrogen-fueled generator or fuel cell. It is also feasible to demand that the tank level at the end of the year equal or exceed the beginning tank level. The storage of hydrogen as a compressed gas in long-lasting tanks is considered in this study. Table 6 details the parameters of the chosen hydrogen tank.

Table 6 Specifications of Hydrogen Tank

Parameter	Value	References
Capital Cost (\$/kgH ₂) installed	500	[28]
O&M Cost(\$/kgH ₂)	0	[29]
Replacement Cost(\$/kW)	500	Estimated
Lifetime (years)	25	[30]

H. Fuel Cells

Fuel Cell systems are extremely clean, emit no pollutants, and have a high efficiency. Hydrogen is the main fuel for fuel cell systems. With the help of an oxidant, such as oxygen from the air, the energy stored in the hydrogen fuel is directly converted into electricity. The proton exchange membrane fuel cell (PEMFC) was chosen for this study because it has a consistent performance under imbalanced hydrogen supply. The PEMFC, which is widely utilized in industrial applications and has a fast dynamic response [31], was selected for this work. Table 7 describes the fuel cell's input requirements.

Table 7 Specifications of Fuel Cell

Parameter	Value	References
Type	PEMFC	Estimated
Capital Cost(\$/kW)installed	2000	[32]
O&M Cost(\$/kW/op.hr)	0.010	[33]
Replacement Cost(\$/kW)	2000	Estimated
Efficiency(%) at rated power	50	[33]
Lifetime (h)	30000	[33]

III. RESULTS AND DISCUSSIONS

A. Case Study

Adrar is an Algerian province in the Sahara region of the country's southwest. Adrar's location is depicted in Figure 1. HOMER is used to extract the necessary climatic data for Adrar from the NASA database (from 1983 to 2005). The wind speed data for the site was obtained from NASA Surface Meteorology and it is depicted in Figure 2. The average annual Wind Speed is: 6.35 (m/s)



Fig. 1 The geographic location of Adrar



Fig. 2 Annual Wind Speed in Adrar

The clearness index, which measures the proportion of solar energy that passes through the atmosphere and strikes the ground, is also depicted in fig. 3. It has a high value in bright, sunny situations and a low value in cloudy weather. Solar scaled average is: 5.78 (kWh/m²/day)

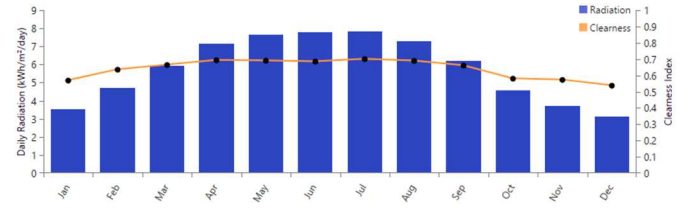


Fig. 3 Annual Solar Irradiation (GHI) in Adrar

The temperature data for the site was also imported from NASA Surface Meteorology and Solar Energy database, as presented in Figure 4. The average annual Temperature is: 24.78 (°C)



Fig. 4 Annual Temperature in Adrar

A load dataset for a drilling camp in Adrar, with a daily AC load of 2426.45kWh and a peak load of 405.71 kW provided by CDER (Renewable Energy Development Center), has been used. The load profile is shown in Fig. 5. The overall demand is higher in the summer months of June-September, with the peak load occurring in August and the least average load occurring in January.

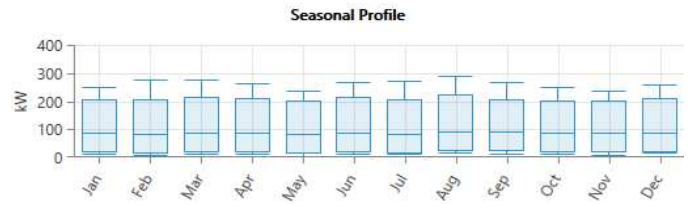


Fig. 5 Annual Load Profile for the microgrid in Adrar

B. Structure of the considered microgrid

The microgrid system under consideration comprises solar photovoltaic (PV) panels, a lithium-ion battery-powered battery energy storage system (BESS), a wind turbine, a diesel generator, and a power to hydrogen system (P2H) that includes an electrolyzer, a hydrogen storage tank, and a fuel cell. The system is depicted in Figure 6.

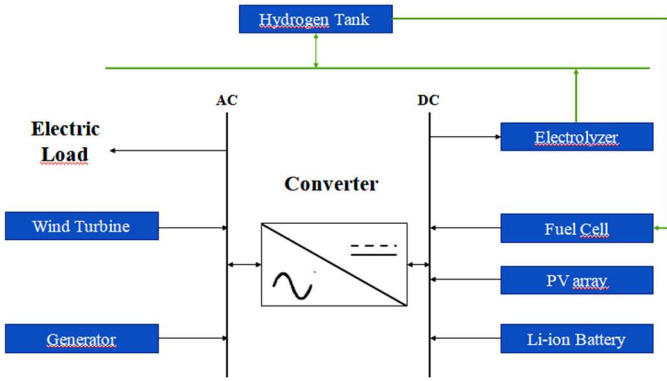


Fig. 6 the considered Microgrid

For this study, two scenarios for standalone microgrid operation are anticipated, each with various configurations. Different combinations were evaluated over a 20-year project lifetime using HOMER Pro to determine the most optimal mix of energy technologies to suit the microgrid's load requirements by comparing the energy balance, economics, and emissions of these scenarios.

Scenario 1: Hybrid Renewable Energy System without hydrogen components.

Scenario 2: Hybrid Renewable Energy System with Power to Hydrogen arrangement.

C. Optimization Results for each scenario

Tables 8 and 9 show the simulation results for each scenario based on the entered inputs. The configuration DG/PV/WT/LI has the least NPC on the first scenario 1. The configuration DG/PV/WT/HS has the least Net Present Cost (NPC) on the first scenario 2.

Table 8 Results of scenario1

Criteria	NPC (\$)	COE(\$)	OC (\$/y)	ICC(\$)	RF%
DG/PV/WT/LI	674628	0.0658	22466	414464	81.9
DG/PV/WT	1.01M	0.0986	53965	386581	63.5
DG/PV/LI	1.61M	0.157	62402	889055	55.6
DG	2.64M	0.258	208856	225000	0
DG/PV	2.65M	0.259	208492	240534	0

Table 9 Results of scenario2

Criteria	NPC (\$)	COE(\$)	OC (\$/y)	ICC(\$)	RF%
DG/PV/WT/HS	1.21M	0.118	9112	1.11M	90.8
DG/PV/LI/WT/HS	1.23M	0.120	14112	1.07M	88.1
DG/PV/LI/HS	4.10M	0.400	19136	3.88M	92.6
DG/PV/HS	4.28M	0.417	40375	3.81M	80.5

Table 10 shows a comparison between the two optimal configurations for each scenario.

Scenario 2 also has the highest Net Present Cost evaluated over the project lifetime of 20 years. Moreover, it is also the highest in emissions due to diesel fuel consumption. On the

other hand, Scenario 1 is the most optimal solution since it has the lowest Net Present Cost than the second scenario and the lowest emissions.

The previous table and results show that the best solution is based on the wind speed and fuel price since its Configuration is based on a wind turbine and diesel generator.

Table 10 Simulation results for the two scenarios

Results		Scenario 1	Scenario 2
System Component	Solar PV capacity (kW)	X	X
	DG1 capacity (kW)	450	450
	Li-ion battery capacity (kWh)	2x100	X
	Fuel cell capacity (kW)	X	250
	Electrolyzer capacity (kW)	X	100
	H ₂ tank capacity (kg)	X	1000
	Converter capacity (kW)	152	114
Economics	Total Net Present Cost (\$)	674628	1.21M
	Cost of Energy (20 years) (\$/kWh)	0.0658	0.118
	Initial Capital Cost (\$)	414464	1.11M
	Operational Cost (\$/yr)	0.166935	0.015775
Electrical	Renewable Energy fraction (%)	81.9	86.0
	Capacity shortage (kWh/yr)	0.02	0.02
	Capacity shortage (%)	0.02	0.02
	Excess electricity (kWh/yr)	7608465	9739868
	Excess electricity (%)	89.4	86.0
	Dispatch strategy	CC	CC
	Total electricity production (kWh/yr)	8507096	11319982
Diesel	DG1 operating (hours/yr)	853	1004
Fuel Cell and Electrolyzer	Fuel cell operating (hours/yr)	X	2,127
	Fuel cell capital cost (\$)	X	2000
	Fuel cell H ₂ fuel (kg)	X	13,975
	Electrolyzer Capital Cost (\$)	X	\$150,000.00
H ₂ storage tank	H ₂ tank autonomy (hr)	X	330
Battery	Battery autonomy (hr)	1.58	X
	Battery usable capacity (kWh)	160	X

IV. CONCLUSION

This work was concerned with using a hybrid PV/diesel system with batteries to achieve cost-effective systems that decrease greenhouse gas emissions as compared to a standalone diesel system. The hybrid PV/diesel system has great potentials and should be developed further for use in power production for distant loads or villages. The production of hydrogen from renewable energy sources could be the solution to long-term energy storage and provide an energy vector with a variety of uses and excellent environmental characteristics. In particular, hydrogen may be transformed into synthetic methane using carbon dioxide from anaerobic digester facilities, which could then be fed into natural gas pipelines and utilized in heating and fuel cell vehicle

applications with almost zero emissions. In the light of the obtained results and observations, increasing the supply from renewable energy sources would allow replacing carbon-intensive energy sources and significantly reducing global warming emissions. In addition to jobs and economic benefits, the use of renewable energy creates stable energy prices. The management methods used in hybrid system depend on the objective of each study. Most researchers focus on techno-economic objectives of hybrid renewable energy system because this type of analysis guarantees technical benefits (increase lifetime, satisfy the load, improve performance) and minimize economical impacts (minimize cost of system, improve saving cost, reduce energy cost).

REFERENCES

- [1] Renewables 2017 Global Status Report, Paris, 2017.
- [2] IEA, Renewables Information-Key Renewables Trends, 2016.
- [3] F. Chellali et al., "A Contribution to the actualization of the wind map of Algeria", *Renewable and Sustainable Energy Reviews*, Elsevier, Vol. 15, Issue 2, 2011, PP- 993-1002.
- [4] L. Ferrer-Martía, A. Garwood, J. Chiroque, B. Ramirez, O. Marcelo, M. Garfi, E. Velo, Evaluating and comparing three community small-scale wind electrification projects. *Renewable Sustain. Energy Rev.* 16 (7), 2012, pp: 5379–5390.
- [5] MINISTERE DE L'ENERGY. Bilan Energetique National Année 2018. Algeria: 2019. <https://doi.org/https://www.energy.gov.dz/?article=bilan-energetique-national-du-secteur>.
- [6] E. Uzlu, Application of Jaya algorithm-trained artificial neural networks for prediction of energy use in the nation of Turkey. *Energy Sources, Part B Econ Planning, Policy* 2019; 14: 183–200. <https://doi.org/10.1080/15567249.2019.1653405>.
- [7] C. Mokhtara, B. Negrou, N. Settou, A. Gouareh, B. Settou, M. A. Chetouane, Decision-making and optimal design of off-grid hybrid renewable energy system for electrification of mobile buildings in Algeria: case study of drilling camps in Adrar. *Alger J Env Sc Technol* 2020.
- [8] A. S. Aziz, M. F. Naim Tajuddin, M. R. Adzman, M. F. Mohammed, MAMM. Ramli, A. Saleh, et al. Feasibility analysis of grid-connected and islanded operation of a solar PV microgrid system: A case study of Iraq. *Energy* 2020; 191:116591. <https://doi.org/10.1016/j.energy.2019.116591>.
- [9] C. Mokhtara, B. Negrou, N. Settou, B. Gouareh, B. S. Abderrahmane, Pathways to plus-energy buildings in Algeria : design optimization method based on GIS and multi-criteria decision making. *Energy Procedia* 2019;162:171–80. <https://doi.org/10.1016/j.egypro.2019.04.019>.
- [10] B. Settou, N. Settou, B. Gouareh, B. Negrou, C. Mokhtara, D. Messaoudi, GIS-Based Method for Future Prospect of Energy Supply in Algerian Road Transport Sector Using Solar Roads Technology. *Energy Procedia* 2019; 162:221–30. <https://doi.org/10.1016/j.egypro.2019.04.024>.
- [11] T. Salameh, C. Ghenai, A. Merabet, M. Alkasrawi, Techno-economical optimization of an integrated stand-alone hybrid solar PV tracking and diesel generator power system in Khorfakkan, United Arab Emirates. *Energy* 2019;190:116475. <https://doi.org/10.1016/j.energy.2019.116475>.
- [12] W. C. Yeh, M. F. He, C. L. Huang, S. Y. Tan, X. Zhang, Y. Huang, et al. New genetic algorithm foreconomic dispatch of stand-alone three-modular microgrid in DongAo Island. *Appl Energy* 2020;263. <https://doi.org/10.1016/j.apenergy.2020.114508>.
- [13] M. Das, M. Anil, K. Singh, A. Biswas, MAK. Singh, et al. Techno-economic optimization of an off-grid hybrid renewable energy system using metaheuristic optimization approaches –Case of a radio transmitter station in India. *Energy Convers Manag* 2019; 185:339–52. <https://doi.org/10.1016/j.enconman.2019.01.107>.
- [14] A. Buonomano, F. Calise, MD. d'Accadia, M. Vicidomini, M. Dentice, M. Vicidomini. A hybrid renewable system based on wind and solar energy coupled with an electrical storage: Dynamic simulation and economic assessment. *Energy* 2018; 155:17489. <https://doi.org/10.1016/j.energy.2018.05.006>.
- [15] AM. Patel, SK. Singal. Optimal component selection of integrated renewable energy system for power generation in stand-alone applications. *Energy* 2019;175:481–504. <https://doi.org/10.1016/j.energy.2019.03.055>.
- [16] JD. Rivera-Niquepa, PM. De Oliveira-De Jesus, JC. Castro-Galeano, D. Hernández-Torres. Planning stand-alone electricity generation systems, a multiple objective optimization and fuzzy decision making approach. *Heliyon* 2020; 6:e03534. <https://doi.org/10.1016/j.heliyon.2020.e03534>.
- [17] K. Anoune, M. Bouya, A. Astito, A. Ben. Abdellah. Sizing methods and optimization techniques for PV-wind based hybrid renewable energy system: A review. *Renew Sustain Energy Rev* 2018; 93:652–73. <https://doi.org/10.1016/j.rser.2018.05.032>.
- [18] A. Felix. Farret and M. Godoy Simões. Micropower system modeling with homer. *Integration of Alternative Sources of Energy*, pages 379–418, 2006.
- [19] J. A. Duffie and W. A. Beckman. *Solar Engineering of Thermal Processes* 2nd ed. Wiley, 1991.
- [20] Abdelmadjid RECIUI and Khaled DASSA, "Design of Standalone Micro-Grid Systems Using Teaching Learning Based Optimization", *Algerian Journal of Signals and Systems*, Vol.2, Issue2, June 2017. pp:75-85.
- [21] MA. Mohamed; AM. Eltamaly; AI. Alolah et al. A novel framework-based cuckoo search algorithm for sizing and optimization of grid-independent hybrid renewable energy systems. *Int J Green Energy*. 2018, 1-15.
- [22] F. Z. Dekhandji, M. Bouaraki, N. Zaidi and A. Recioui, "Demand Response and Optimal Sizing of Hybrid Renewable Energy Systems", *International Journal of Control, Energy and Electrical Engineering (ICEEE)*, Vol. 7, pp:35-40.
- [23] V. Suresh, M. Muralidhar, and R. Kiranmayi. Modelling and optimization of an off-grid hybrid renewable energy system for electrification in a rural areas. *Energy Reports*, 6(1):594–604, 2020.
- [24] S. Singh.; M. Singh.; S. Chandra. Feasibility study of an islanded microgrid in rural area consisting of PV , wind , biomass and battery energy storage system. *Energy Convers Manag*. 2016. 128:178-190.
- [25] M. Smaoui; A. Abdelkafi; L. Krichen. Optimal sizing of stand-alone photovoltaic/wind/hydrogen hybrid system supplying a desalination unit. *Sol Energy*. 2015.120: 263-276.
- [26] S. Ericson and D. Olis. A comparison of fuel choice for backup generators. *NREL/TP-6A50-72509*, 2019.
- [27] A. Recioui and H. Bentarzi, Optimizing smart grid operation and control, *IGI Global*, 2021.
- [28] AS. Jacob; R. Banerjee; PC. Ghosh. Sizing of hybrid energy storage system for a PV based microgrid through design space approach. *Appl Energy*. 2018.212: 640-653. DOI: 10.1016/j.apenergy.2017.12.040
- [29] Y. Zhang; PE. Campana; A. Lundblad; J. Yan. "Comparative study of hydrogen storage and battery storage in grid connected photovoltaic system: Storage sizing and rule-based operation," *Applied Energy*, Elsevier, vol. 201(C), pages 397-411.
- [30] J. Hinkley ; J. Hayward; R. McNaughton; RG. Csiro. Cost assessment of hydrogen production from PV and electrolysis. 2016.
- [31] Y. Kebbat and L. Baghli. Design, modeling and control of a hybrid grid-connected photovoltaic-wind system for the region of Adrar, Algeria. *Int. J. Environ. Sci. Technol.* (2022). <https://doi.org/10.1007/s13762-022-04426-y>
- [32] DN. Luta; AK. Raji. Decision-making between a grid extension and a rural renewable off-grid system with hydrogen generation. *Int J Hydrogen Energy*. 43, 2018. 1-14.
- [33] K. Dassa, and A. Recioui, Application of Optimization to Sizing Renewable Energy Systems and Energy Management in Microgrids: State of the Art and Trends. In: *Applications of Nature-Inspired Computing in Renewable Energy Systems*, IGI Global, 2022. pp:60-94.